

Northfield Credibility Studio

Brief technical report | 16 September 2026

Assessing the credibility of organisational knowledge from fragmented data

1. Project overview

This system implements the iLab 14-01 group proposal: a credibility assessment pipeline for Northfield Software Ltd (Decidr partner dataset). Organisational knowledge is scattered across policies, process documents, interviews, meetings, Slack communications, CRM records, project tasks, and audit logs. The pipeline retrieves evidence for a claim, weighs sources by authority, scores credibility, and returns an inspectable verdict, including Conflicted when formal rules and practised behaviour disagree.

2. Data preparation

Phase 1 and Phase 2 Northfield data packs were combined into one working dataset. Phase 2 is the superset. Documents with the same ID in both phases are deduplicated (Phase 2 kept) to avoid double-counting evidence.

- Combined raw folder: data/combined/ (45 files + labelled sample claims)
- Pipeline output: data/outputs/chunks.jsonl (743 searchable text chunks)
- Gold claims: data/gold/gold_claims.json (25 human-scored test claims)
- Rulebook: rulebook/ (process clusters P1-P8 and dispute-type guidance)

3. Backend pipeline (eight stages)

Stage 1 - Ingest

Reads Phase 1 + Phase 2 markdown and CSV files. Splits markdown on headings; one chunk per CSV row. Writes documents.jsonl and chunks.jsonl.

Stage 2 - Extract

Rule-based claim extraction using modal verbs (must, shall, required) and role names. Outputs unreviewed candidate claims.

Stage 3 - Retrieve

BM25 keyword search finds top-k evidence chunks. Rulebook must-fetch IDs are injected so linked documents are not missed.

Stage 4 - Reason

Each chunk is labelled support, contradict, or unrelated via lexical overlap and negation cues. Unrelated chunks are dropped.

Stage 5 - Weight

Authority tier applied (policy 1.0 down to comms 0.35). Superseded/draft penalties. Duplicate Slack copies collapse to one independent source.

Stage 6 - Score

Bayesian log-odds update from weighted evidence produces a 0-1 score and status label.

Stage 7 - Label

Batch run on gold and labelled sample claims. Writes labelled_claims.jsonl.

Stage 8 - Explain

Deterministic 6-step thought-process trace plus narrative. Optional LLM rewrites prose only; it never changes the score.

4. Rulebook and weightage

The Northfield Rulebook encodes client guidance. Four dispute types determine which sources win:

- RULE: approved in-force policy beats stale/draft documents
- HAPPENED: CRM and audit log beat Slack memory for recorded events
- PRACTICE: meetings and interviews; copied Slack threads count as one source
- GAP: if no owner is formally assigned, verdict is Unsupported (gap)

Process clusters (P1-P8) link related document IDs. Must-fetch documents are boosted during retrieval and weighting. Formal and practised claims are scored separately, never averaged.

5. Verdict labels

Supported (score ≥ 0.75) | Likely (≥ 0.6) | Conflicted (weight on both sides) | Unlikely (≤ 0.4) | Unsupported (≤ 0.25) | Not enough evidence. Confidence (High/Medium/Low) reflects evidence completeness, separate from score direction.

6. Streamlit UI

The dashboard uses a black background and orange accent (#FF6A12), matching the reference design. Launch with: `streamlit run app.py`

Overview

Introduction, eight-stage method strip, corpus KPIs, usage guide.

Command centre

KPI dashboard: counts, source-type charts, circular gauges, bar/line charts.

Data atlas

Browse ingested documents and chunks by source type and phase.

Claim workbench

Score claims; shows verdict, thought-process trace, evidence table.

Evidence chat

Ask questions with retrieved chunk citations.

Claim extraction

View rule-based candidate claims from the corpus.

Evaluation

Compare verdicts against gold expected statuses and retrieval recall.

Method

Authority weights, status thresholds, pipeline documentation.

7. How to run

```
cd ~/Documents/northfield_credibility_studio
source .venv/bin/activate
python pipeline/run_all.py
streamlit run app.py
```

Optional: set `OPENAI_API_KEY` for LLM explanations (score remains deterministic).

8. Folder layout

```
pipeline/  01_ingest, 02_extract, 07_label, run_all.py
src/       retrieve, reason, weight, score, explain, rulebook
```

Northfield Credibility Studio | iLab 14-01 / Decidr

services/ engine.py (orchestrates pipeline for UI)
ui/ black + orange theme
data/outputs/ chunks.jsonl, labelled_claims.jsonl

Team: iLab 14-01 - Decidr x Northfield Software Ltd | Assessment date: 31 July 2026